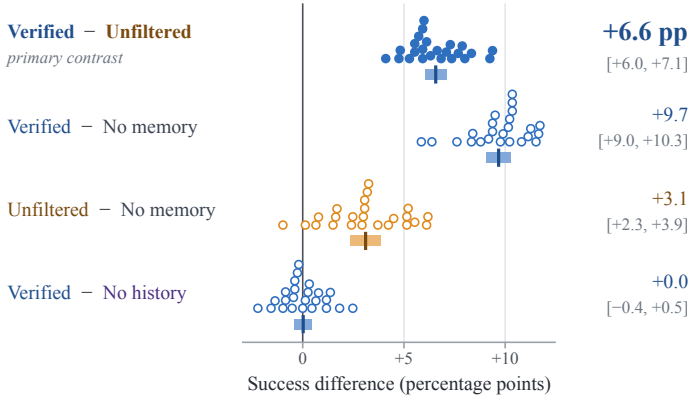
Verified — checked + 25% history
No history — checked records only
Unfiltered — unchecked records
No memory — nothing stored
SYNTHETIC DATA
constructed demonstration, not results from real agents

(a) Paired success differences

dot: one of 24 matched seed blocks; bar and tick: 95% CI and mean



(b) Gain by task and budget

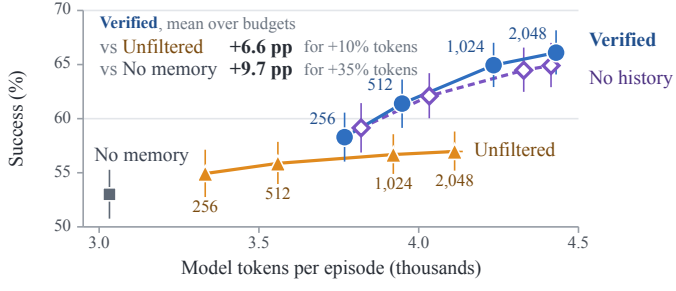
Verified − Unfiltered, percentage points
95% CI includes 0

Memory budget (tokens)

	256	512	1,024	2,048	mean
Fact recall	+0.3	+0.5	+1.3	+1.9	+1.0
Code repair	+4.1	+6.9	+9.9	+11.1	+8.0
Tool routing	+5.4	+8.6	+11.6	+12.5	+9.5
Long horizon	+3.6	+6.0	+10.2	+11.1	+7.7
mean	+3.4	+5.5	+8.3	+9.1	**+6.6**

(c) Success versus token cost

mean over 24 seed blocks; bars: 95% CI; numbers on points: memory budget (tokens)



(d) Rejection-history diagnostic

